165 (2022) 207205

Contents lists available at ScienceDirect

Micro and Nanostructures

journal homepage: www.journals.elsevier.com/micro-and-nanostructures

Excitation density dependent carrier dynamics in a monolayer MoS_2: Exciton dissociation, formation and bottlenecking

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A B S T R A C T

Research on Molybdenum Disulfide (MoS_2) monolayer at high excitation densities is critical due to its rising demand in optoelectronic applications such as lasers, compact optical parametric amplifiers, and high power detectors. We show that at high excitation densities, measurements on a single monolayer MoS_2 reveal a transient response rich in physical processes. Despite the fact that carriers are excited straight to the A-excitonic state, the time and fluence dependence of the observed optical response indicates exciton dissociation followed by hot-carrier generation. In the picosecond time scale, these hot-carriers form excitons, which have a longer lifetime at higher excitation densities, implying a bottlenecking process. For the use of MoS_2 at high-excitation densities, the fluence dependency of the relaxation mechanisms detailed here is essential.

1. Introduction

Understanding the ultrafast carrier dynamics and the accompanying physical processes in Molybdenum Disulfide (MoS_2) is vital for its applications in electronics, spintronics, valleytronics, and quantum technologies [1–5]. Various groups have already documented the use of MoS_2 in photodetectors, solar cells, lasers, and optical parametric amplifiers (OPA) [6–9]. Due to strong quantum confinement and Coulomb interaction, the excitons in monolayer (ML) MoS_2 have a high binding energy of roughly 0.5 eV [10–12]. The fact that ML MoS_2 exhibits significant light-matter interaction in the visible area through three excitonic transition peaks, A, B, and C, exemplifies such optoelectronic applications [13]. The A-exciton has the lowest transition energy among these three excitons. As a result, any carriers excited in the ML MoS_2 soon relax to the A-exciton state at ambient temperature, resulting in intense photoluminescence [13,14].

At various excitation conditions, several groups have investigated the ultrafast carrier dynamics in ML MoS_2. Several relaxation and scattering mechanisms occur, including exciton relaxation via defect states [15,16], Auger recombination and defect states assisted Auger recombination [17–20], exciton-exciton annihilation [21–25], and carrier cooling by phonon emission [26–30]. The contribution of any given physical process to the optical response depends strongly on the excited carrier density. For a thorough knowledge of the material, it is necessary to measure the dependence of various time constants on excitation densities and comprehend the related physical processes. Fluence-dependent studies can also provide information about the underlining physical process. For example, the efficiency of Auger scattering is known to increase with the density of excited carriers [19,20]. Hence, a reduction in the decay time is expected with an increase in excitation fluence in case of Auger recombination. On the other hand, the lifetime of an excited carrier in a given state can increase due to a bottlenecking process, if the number of available states to which it is decaying is getting reduced as

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https://doi.org/10.1016/j.micrna.2022.207205
Received 25 November 2021; Received in revised form 7 March 2022; Accepted 17 March 2022
Available online 28 March 2022

carriers occupy these states [14,31]. It is also possible that at high excitation densities more number of phonons are generated in the sample which gets reabsorbed by the carriers preventing it from decaying faster [31,32]. This processes will leads to an increase in the decay time of carriers [14,31,32].

In this article, we investigate ultrafast carrier dynamics in ML MoS_2 at high excitation fluences ranging from 1.23 $mJcm^{-2}$ to 7.2 $mJcm^{-2}$, which translates to high excitation carrier densities of $\sim 6.6 \times 10^{13}$ cm^{-2} to 4×10^{14} cm^{-2} by exciting and probing the carriers at A-exciton transition energy i.e., 1.84 eV (672 nm) with a spectral width of 0.072 eV. Though the carriers are excited directly to the A-exciton state by the pump pulse, the fluence and time dependence of transient absorption show that the excitons are getting dissociated forming free carriers. A part of these carriers relax to lower energy states while some of them form hot-carriers through the Auger scattering. It will also be demonstrated that at higher excitation fluences, these carriers relax to produce longer-lived A-excitons, indicating a bottlenecking process. Such ultrafast carrier dynamics studies at high excitation densities are relevant in the case of MoS_2 for its applicability in devices that handle high carrier densities [6–9].

2. Results and discussion

The ML MoS_2 sample under study was grown by chemical vapor deposition technique on a sapphire substrate [33]. The optical microscopic investigation of the sample shows several individual and connected flakes covering the substrate. Atomic Force Microscopic (AFM) measurement was performed on a selected flake to measure its thickness and is shown in Fig. 1(a). The thickness of the MoS_2 flake estimated from the AFM image was $\sim$ 0.8 nm, showing that the selected flake is a ML [34,35]. Using a microscope, a spectrometer, and a white light source, we have measured the absorption spectrum of the same flake and is shown in Fig. 1(b). The absorption spectrum shows three excitonic peaks, A, B, and C, at 1.84 eV, 1.98 eV, and 2.87 eV, respectively, which are expected of a ML MoS_2 [13,17]. The transient transmission measurements were performed using an OPA operated at 1.84 eV (near the A-exciton peak). The OPA was pumped by an 800 nm, 35 fs oscillator-amplifier operated at 1 kHz. These transient measurements were performed in a standard degenerate pump-probe configuration [34,36]. Using off-axis parabolic mirrors, the pump and probe beams were focused on the same selected ML flake while viewed through an optical microscope. The pump and probe beams have a Gaussian spatial profile and have a full width at half maximum of about 72 μm and 24 μm respectively. Since the selected ML flake on which the measurements were performed is much larger than the probe beam, only a single ML MoS_2 is actually probed. Fig. 1(c) shows the optical image of the selected flake, and the dotted circle marks the area at which the transient measurements were performed. The pulse width of the OPA output was measured to be 60 fs at the sample location. Due to the short temporal width of the pulses, the spectral width of the laser pulse is about 0.072 eV. The photo-detector integrates the energy in the entire pulse. Thus, the changes in the transmission at 1.84 eV integrated over a width of 0.072 eV are monitored by the measurement. Measurements were performed at different pump fluences in the range of 1.23 $mJcm^{-2}$ to 7.2 $mJcm^{-2}$. The corresponding density of carriers excited in the sample was estimated to be in the range of 6.6×10^{13} cm^{-2} to 4×10^{14} cm^{-2}, [31,37]. To ensure the stability of the flakes during the transient measurements, we have systematically exposed several monolayer MoS_2 flakes to fluences up to $\sim$ 100 $mJcm^{-2}$. Both Raman and AFM topography measurements on these exposed flakes show that they were stable up to 50 $mJcm^{-2}$, [34].

In Fig. 2, we show the measured transient transmission ($\Delta T/T$) through the ML MoS_2 sample with respect to the delay between the pump and probe pulses at various pump fluences (note that we have plotted $-\Delta T/T$ in the y-axis). The transient transmission, $\Delta T/T$, stands for $(T' - T_0)/T_0$, where T_0 and T' are the transmissions of the probe beam through the sample in the absence and presence of the pump beam, respectively. With the arrival of the pump pulse, the transmission through the sample starts to reduce. As the delay between the pump and the probe pulse increases, this negative change in transmission reaches a maximum, nearly within the pulse

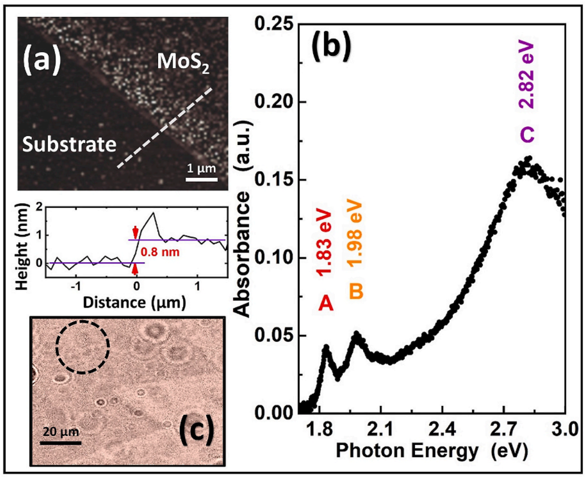

Fig. 1. (*a*) AFM topography image of the MoS_2 flake on which the transient transmission measurements were performed. The bottom plot shows the height variation along the dotted line. (*b*) The measured microscopic absorption spectrum of the same flake. (*c*) The optical image of the flake with the circle marking the location at which transient measurements were carried out.

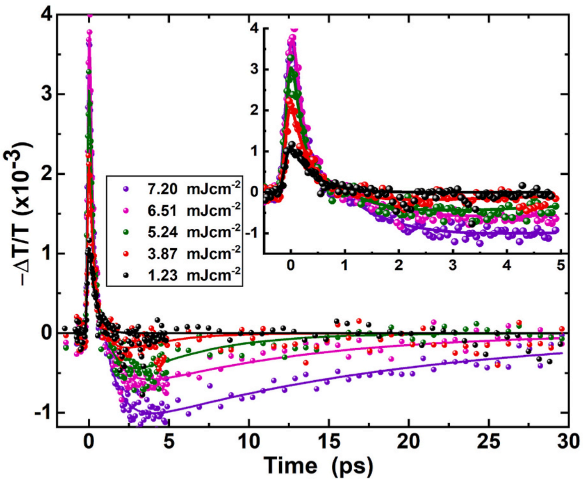

Fig. 2. Fluence dependent transient transmission signal of the selected ML MoS_2 flake at A-exciton peak (note the negative sign of y-axis). Inset shows the same plot zoomed over the first few picosecond time scale. The corresponding color lines are the best fit to the data using Eq. (1).

duration. With further increase in delay, the change in transmission starts recovering. For a fluence of 1.23 $mJcm^{-2}$ the transmission through the sample recovers fully ($\Delta T = 0$) within the first few picoseconds. As the pump fluence increases, the magnitude of the initial negative peak also increases. At higher fluences, the ΔT recovers to zero, but it changes sign to become positive at later times. A positive maximum is reached at nearly 3 ps. Upon further delay, this positive signal also decays nearly exponentially to zero in tens of picoseconds.

To investigate the carrier dynamics further, we plot the dependence of the measured magnitude of the initial negative peak, $|\Delta T/T|_{neg}$, on the pump fluence in Fig. 3(a). A linear fit to the data clearly depicts that the $|\Delta T/T|_{neg}$ increases linearly with the pump fluence. Fig. 3(b) shows the pump fluence dependence of the magnitude of the later positive peak in $\Delta T/T$, $|\Delta T/T|_{pos}$, which occurs nearly at 3 ps delay. The $|\Delta T/T|_{pos}$ remains nearly negligible at lower fluences but starts to increases quite rapidly with increasing fluence. A best fit to the data with a quadratic function, given by $y = ax^2$, with a as a fitting parameter, is also shown in Fig. 3(b). Clearly $|\Delta T/T|_{pos}$ has a strong quadratic dependence on fluence. Based on these observations, we find that there are two different processes which contribute to the overall temporal variations in the observed time dependence of $\Delta T/T$. First, an instantaneous process which contributes negatively to $\Delta T/T$ and it depends linearly on the excitation density. A second delayed process contributes positively to the $\Delta T/T$ and it depends quadratically on the excitation density. To further understand the time dependence of $\Delta T/T$, we fitted the experimental data to an empirical function,

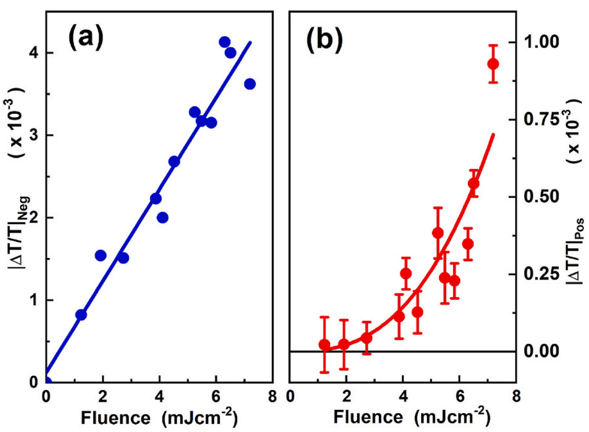

Fig. 3. (*a*) Dependence of the magnitude of the initial negative peak in the time dependence of $\Delta T/T$ with fluence (Fig. 2). The solid line is the best linear fit to the data. (*b*) Fluence dependence of the magnitude of the positive peak which occurs near 3 ps delay. The solid line is the best fit with quadratic dependence (see text for details).

$$\mathscr{F} = \frac{A_0}{2}\left[1+\mathscr{E}\left(\frac{t}{t_0}\right)\right]\exp\left(-\frac{t}{\tau_s}\right) + B_0\left[1-\exp\left(\frac{t-t_d}{\tau_r}\right)\right]\exp\left(-\frac{t-t_d}{\tau_L}\right), \tag{1}$$

where $\mathscr{E}$ is the Gaussian error function, and t_0 is related to the pulse width of the laser. A_0 and τ_s are the amplitude and decay time of the initial negative contribution, respectively. B_0 is the amplitude of the delayed (by t_d) positive peak, with a rise time of τ_r and a decay time of τ_L. The above equation is derived based on the present experimental conditions and observations. The temporal shape of our laser pulse is nearly a Gaussian function. The change in the signal during the pump pulse is expected to depend on the number of carriers present in the excited state. This number at a given time depends on the time integrated intensity of the pump pulse, which is an error function. At the end of the pump pulse, the signal is found to decay nearly exponentially. Based on these two observations, an error function multiplied with an exponential decay function is used to model the initial negative change in transmission. On the other hand, the form of the increase and decay of the latter positive change in transmission suggests an exponentially increasing and decaying function with different time constants for fitting the positive contribution to the $\Delta T/T$ signal. This constitutes the second part of Eq. (1). This function $\mathscr{F}$, given by Eq. (1) was used to fit the experimental data for each fluence. The obtained best fit to the experimental data at each fluences is also shown in Fig. 2.

Fig. 4(a) shows the fluence dependence of the estimated τ_s. At lower fluences, the decay time of the initial negative peak, τ_s, remains nearly at 0.45 ps. As the pump fluence increases, it starts reducing quickly and eventually saturating at about 0.2 ps at higher fluences. We find that the later positive contribution is always time-shifted (t_d) by about 1.1 ± 0.1 ps. Fig. 4(a) also shows the fluence dependence of the estimated rise time of the positive peak, τ_r. Further, since the positive contribution does not appear at lower fluences, τ_r could be only be estimated at higher fluences. Clearly, the τ_r remains more or less constant at around 1 ps. It is important to note that only with the appearance of the positive contribution to the $\Delta T/T$ at higher pump fluences, the τ_s starts to reduce, indicating that they are related.

When carriers are excited with photon energy resonance to the A-exciton peak, the pump pulse primarily excites A-excitons in the ML MoS_2 [32,38]. When the excited carrier densities are in the range of 10^{12} cm^{-2} to 10^{13} cm^{-2}, the transmission through the sample was found to increase, indicating a band filling or band bleaching process [28]. On the other hand, the presence of a large number of excited carriers in the conduction band can lead to a reduction in bandgap, called bandgap renormalization (BGR), and can reduce the transmission at lower photon energies [39,40]. Although nearly only A-excitons are excited by the pump pulse, in the range of excitation densities used here, excitons are known to dissociate within the pulse duration and form free-carriers [26,27,34]. Thus, by the end of the pulse, the abundance of carriers in the conduction band would have led to the BGR reducing the probe transmission [34]. In general, BGR can lead to exciton peak shift and exciton line width broadening [41–44]. Both of these effects can contribute to the observed change in transmission. Note that due to the finite spectral width of the pump and probe pulses, the optical response around A-exciton is integrated in the observed transient transmission. The line width broadening is found to be dominant in presence of exciton-exciton and exciton-phonon interactions [43,44]. Bera et al. showed that when working above Mott density, the creation of electron-hole plasma will lead to a decrease in bandgap energy and hence the red-shifting of exciton peak [41]. In the present case, the fluence levels are such that the excited carrier densities are higher than the Mott density, hence exciton-exciton interactions are expected negligible [23,24]. Thus, we expect the exciton peak shift to contribute more to the observed change in transmission when compared to the broadening. Further, Bera et al. also showed that the exciton peak-shift increases linearly with fluence. In the present case also we find that the magnitude of the peak change in the $\Delta T/T$ increases linearly with fluence (Fig. 3(a)). In view of all the above-mentioned processes, an overall negative peak in transient transmission is reached by the end of the pump pulse.

Let us now look at the origin of the fast recovery of the observed initial negative peak, which has a response time of τ_s. At low excitation densities, it has been shown earlier that the recovery of the initial change in the optical response can be attributed to the

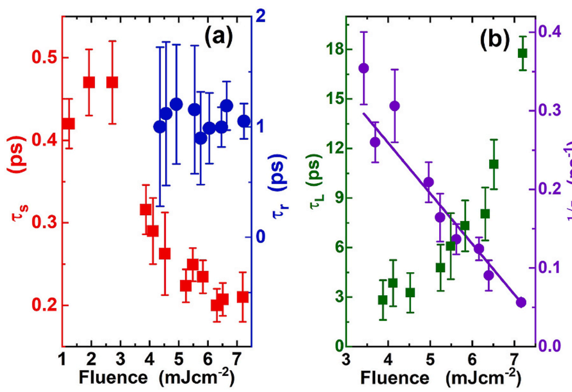

Fig. 4. (a) The fluence dependence of the estimated τ_s (decay time of the initial negative signal) and τ_r (the rise time of the later positive signal). (b) The fluence dependence of the decay time of the positive amplitude, τ_L. The solid line is the linear fit to the $1/\tau_L$ data.

relaxation of carriers to the defect states [14–16]. The decay time of such relaxations is reported to be in the order of 0.2 ps–0.6 ps and is also expected to be fluence independent [14–16,31]. The relaxation of carriers out of the conduction band to the defect states would reduce the effect of BGR, causing the recovery of $\Delta T/T$. In the present case, the recovery time observed at low excitation fluences, i.e., below 2.8 $mJcm^{-2}$ shows a nearly constant recovery time of around 0.45 ps. The density of defects in ML MoS_2 is expected to be of the order of 10^{13} cm^{-2}, [45–47]. Thus, at higher fluence densities, the number of the carriers excited in the ML MoS_2 is expected to be larger than the number of available defect states. Although a partial recovery of the initial peak could be attributed to the relaxation of carriers to the defect states, the observed complete recovery of the initial negative peak cannot be fully explained by decay to the defect states. Further, if the defect states were saturating at higher excitation densities, we should have observed an increase in the recovery time due to the non-availability of states for carriers to decay. This is the opposite of that observed in the experiment (Fig. 4). The presence of a large number of excited carriers can lead to Auger based recombination either directly to the valence band recombination or through defect-assisted process [20,32]. Such Auger based recombination can lift the carriers deep into their respective bands [32]. Such recombination to the valence band and the defect states will reduce the number of excited carriers, reducing the effect of BGR and leading to the recovery of $\Delta T/T$. It is well known that the Auger process is more efficient at higher excitation densities. The reduction in the recovery time, τ_s, at higher fluences indicates that Auger based relaxation is contributing to the reduction of signal. A schematic for explaining these processes which contribute to the initial negative contribution in $\Delta T/T$ is shown in Fig. 5(a).

Let us now discuss the origin of the second positive increase in transient transmission. The Auger scattering not only contributes to the relaxation of some carriers but also excites an equal number of carriers to the higher excited states (Fig. 5(a)) [20,32]. As these carriers relax to the band edge, eventually excitons will start forming again due to lower carrier density as compared to the initial excitation condition [48]. Such formation of A-excitons would lead to bleaching of available states for absorption of the probe, leading to an increase in transmission. The band bleaching will give a positive contribution to the $\Delta T/T$ canceling the negative effect of BGR. Thus, the action of Auger should also be coupled with the bleaching of states at a later time. This explains why the reduction of τ_s is associated with the increase of positive $\Delta T/T$ signal (Figs. 2 and 4). Exciton formation time and hot-carrier relaxation time in ML MoS_2 are reported to be of the order of few hundreds of femtoseconds to a picosecond [48]. Since the build up of carriers, which corresponds to the τ_r, is happening at 1 ps time scale, we attribute this to the exciton formation time in the present case.

Defect states present in the MoS_2 provides a fast relaxation pathway for the A-excitons [23,31]. Thus, the life time of A-excitons will depend also on the number of defect states available for them to relax. If the excitation carrier density is much lower than the defect density, then the relaxation time will remain independent of the excitation fluence [19,29]. On the other hand, if the excited carrier density is higher than the density of defect states, these states will get filled up, reducing the relaxation rate of excitons [14,31]. This bottlenecking process will lead to an increase in the life time of A-excitons. Further, if the excitation density is high enough, a large number of phonons will be emitted by the cooling of hot-carriers produced by Auger [32]. These phonons will be reabsorbed by the carrier in defect states bringing them back to A-exciton state (Fig. 5(b)). This will lead to an increase in the life time of carriers in the A-exciton state, which is also called the bottleneck effect [31,32]. A similar bottlenecking process is also reported in monolayer WS_2, WSe_2, $MoSe_2$, $MoTe_2/WTe_2$ heterostructure [49–51]. As mentioned earlier, the reported maximum defect state density in a CVD grown ML MoS_2 is in the order of $\sim 10^{-13}$ cm^{-2}, [46,47]. In the present case, the excitation carrier densities used are higher than the defect state density. Therefore, defect states will eventually saturate, increasing the life time of A-excitons [31]. Further, the phonon reabsorption will also contribute to the increase in the life time of carriers in A-exciton. Thus the relaxation time, τ_L will increase as the excitation fluence increases (Fig. 4(b)). We find that $1/\tau_L$ depends linearly on the fluence, which is also shown in Fig. 4(b). Thus the scattering process involved depends directly on the number of excited carriers. A schematic of the origin of the later positive contribution to $\Delta T/T$ is shown in Fig. 5(b).

3. Conclusions

Because of the growing need for optoelectronic applications, research into expand TMDC materials at high excitation densities is critical. We have shown that when studying ultrafast carrier dynamics on ML MoS_2 at high excitation densities, the temporal response is considerably different from that seen at lower excitation densities. Fluence independent decay to defect states as well as radiative recombination should have been observed when carriers were excited to a lowest lying level. Despite the fact that carriers are excited directly to the A-excitonic state, the time dependency of the measured optical response indicates a scattering process, which gets more efficient as carrier concentrations increase. The observed variations in the early alterations in the optical response are due to auger scattering of carriers produced by exciton dissociation. A longer relaxation process with increasing decay time suggests a bottlenecking process caused by defect state saturation and re-absorption of available phonons. For the use of MoS_2 at high-excitation densities, the fluence dependence of the relaxation processes detailed here is crucial.

Credit author statement

Durga Prasad Khatua: Investigation, Formal analysis, Resources, Writing – original draft. Asha Singh: Formal analysis, Validation, Resources. Sabina Gurung: Formal analysis, Validation, Resources. J. Jayabalan: Conceptualization, Investigation, Resources, Supervision, Writing – review & editing.

Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

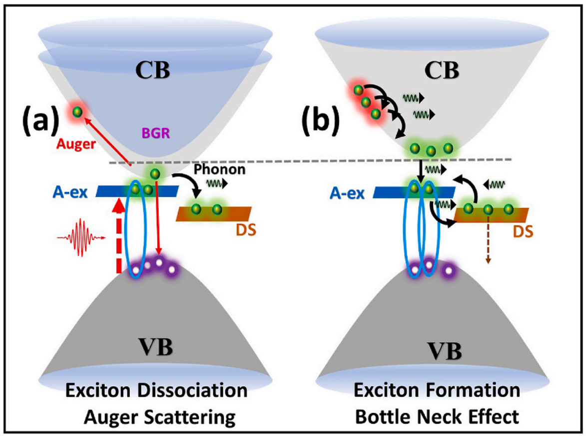

Fig. 5. Schematic of the proposed processes: (*a*) Excitation of A-excitons (A-ex) by pump, exciton dissociation leads to BGR, and creation of hot-carrier by Auger scattering (CB-Conduction band and VB-Valance band). (*b*) Cooling of hot-carriers leading to the formation of exciton and relaxation of carries to the defect states (DS).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

D.P.K. is thankful to RRCAT, Indore for the financial support under HBNI programme. All authors are thankful to Dr. Salahuddin Khan and Vijay Singh Dawar for their valuable inputs during this work. The authors are thankful to Dr. Tarun Kumar Sharma, Head, MSS for his suggestions and support.

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